Unit: Unit 2: Exponents Radicals
Topic: Exponential Equations

Name: ________________
Date: ____________

Homework (Jalapeno)

Q1. Solve for x: $2^x = 16$
(A) $x = 2$
(B) $x = 3$
(C) $x = 4$
(D) $x = 5$
(E) $x = 6$

Q2. The ancient Egyptians used exponential equations to calculate the area of the Great Pyramid. If $3^x = 81$, then x is
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6

Q3. Solve for y: $5^y = 125$
(A) $y = 1$
(B) $y = 2$
(C) $y = 3$
(D) $y = 4$
(E) $y = 5$

Q4. The ancient Greek mathematician Euclid used the concept of exponents to calculate the volume of a sphere. If $2^x = 32$, then x is
(A) 3
(B) 4
(C) 5
(D) 6
(E) 7

Q5. Solve for z: $4^z = 256$
(A) 3
(B) 4
(C) 5
(D) 6
(E) 7

Q6. In ancient Babylon, mathematicians used exponential equations to calculate the area of triangles. If $6^x = 216$, then x is
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6

Q7. The ancient Chinese mathematician Liu Hui used the concept of exponents to calculate the volume of a cylinder. If $8^y = 512$, then y is
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6

Q8. Solve for w: $9^w = 729$
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6

Open Ended Questions (FRQ)

Q9. Solve for x: $2^x + 5 = 21$

Q10. The ancient Mayans used exponential equations to calculate the number of days in a year. If $3^x = 243$, then x is $\boxed{}$

Chef's Tips

- Check for errors in calculation
- Ensure students show their work
- Provide feedback on problem-solving strategies